In this project, I have created a Windows virtual machine in Azure to host a new website for Sam's Scoops. I will deploy the virtual machine for the Sam's Scoops web server using the virtual network Web_Server on a 172.16.1.0 subnet. This virtual machine was named "SamScoopsWeb".

This SOP will guide a through steps as future reference for setup with screenshots that demonstrate the correct actions

Create the network and resource group

Step 1

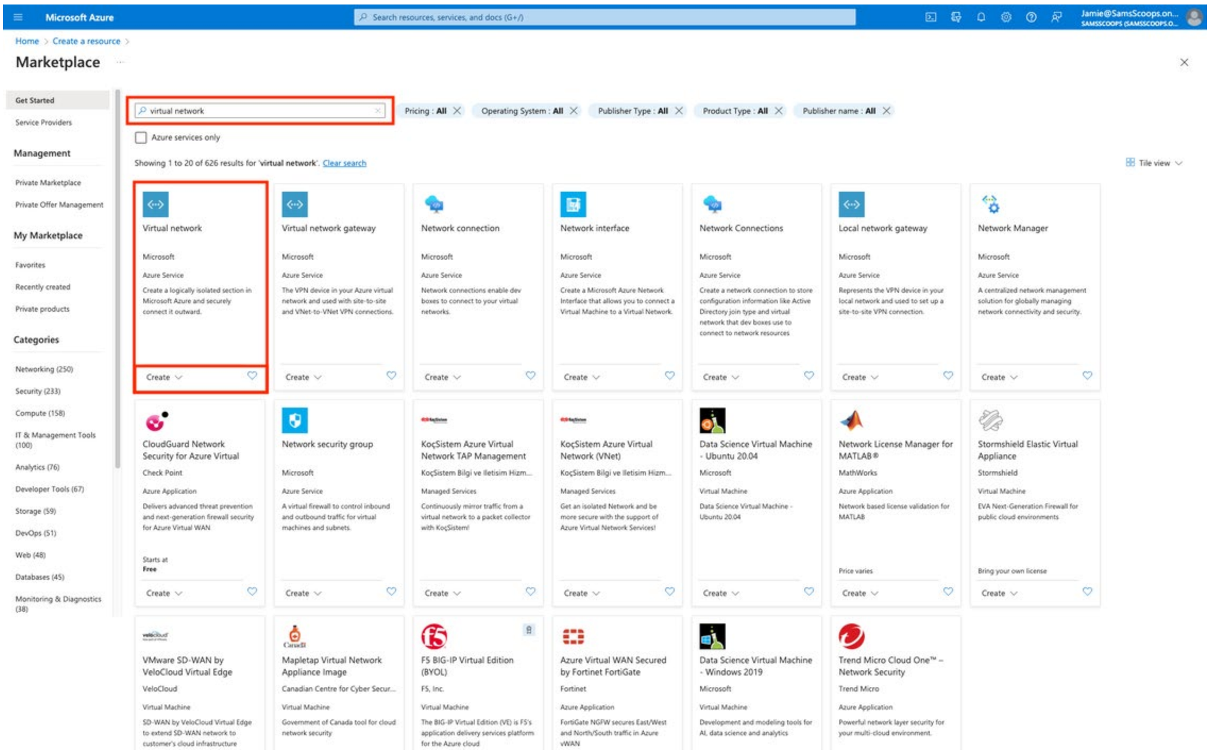
Sign into the [Azure portal](https://portal.azure.com) with your credentials.

Step 2

In the Azure portal menu, select the **Create a resource** button located on the left-hand side of the screen.

Step 3

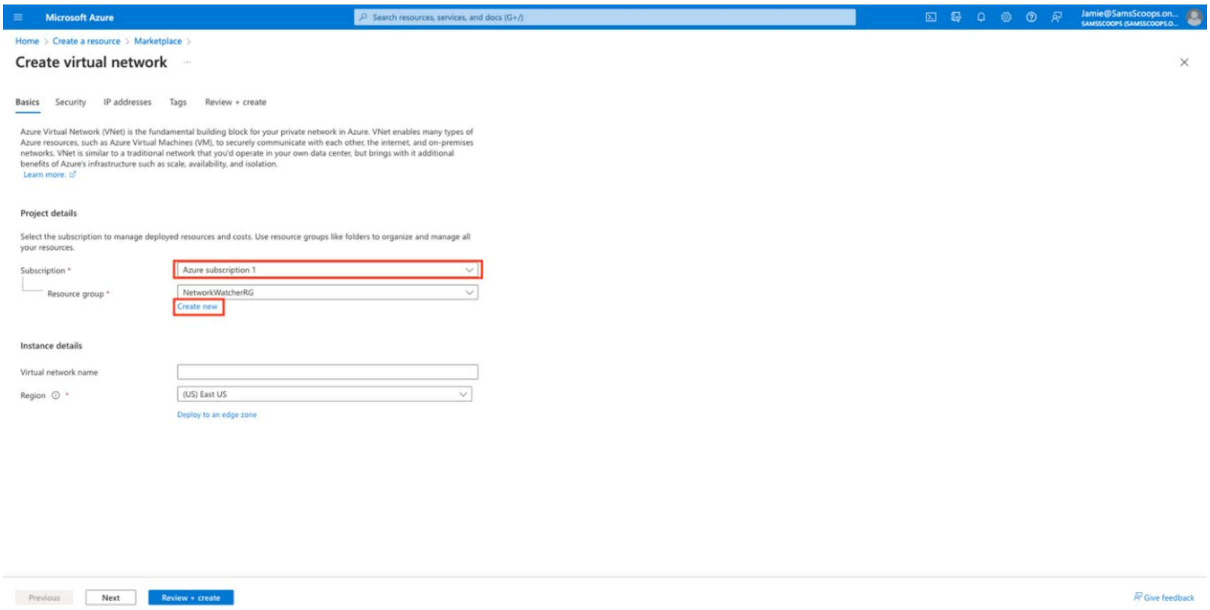
Next, type "Virtual network" in the search bar, select **Virtual network** from the results and select **Create**.

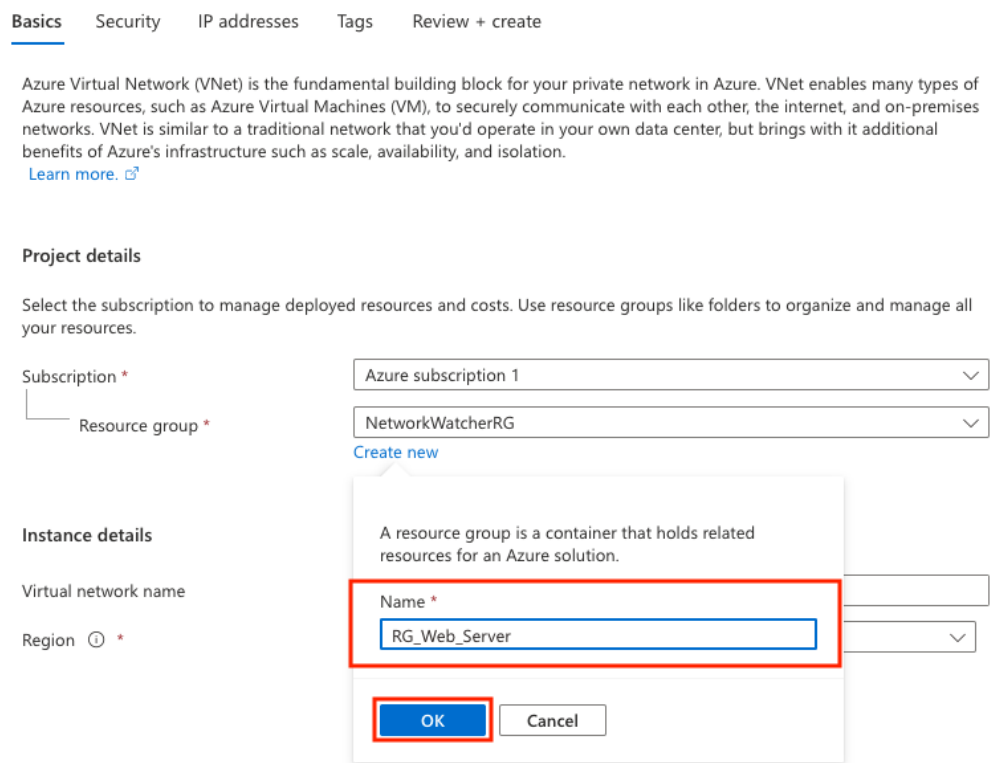


Step 4

In the **Basics** tab of the Create a virtual network wizard, fill out the following information:

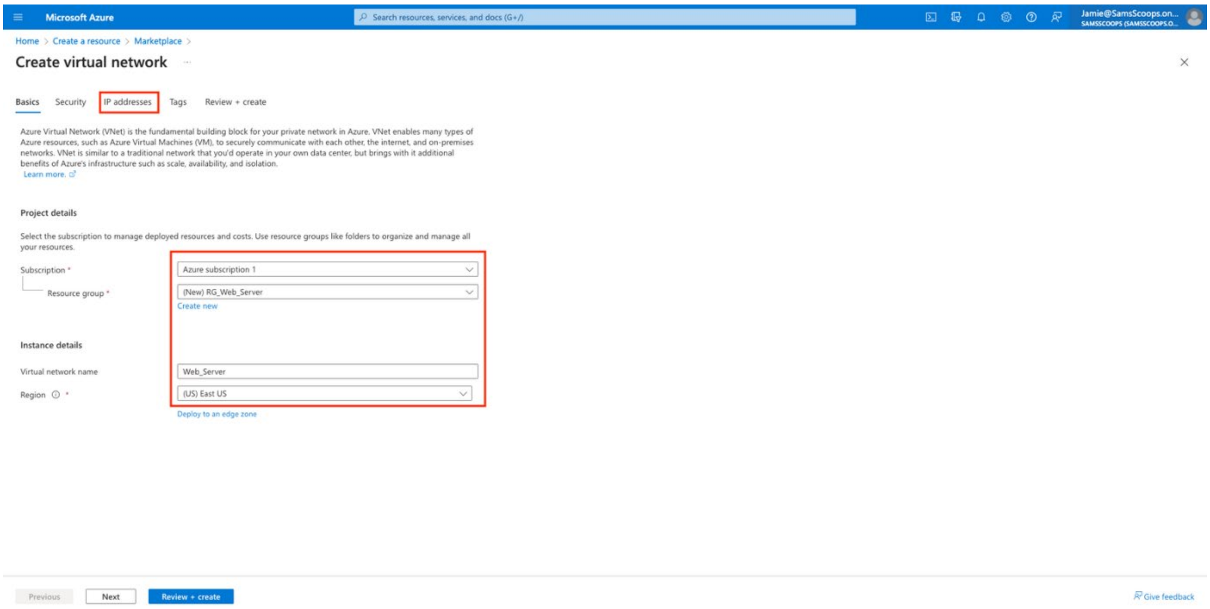
1. Subscription: Choose the subscription that you want to use.
2. Resource group: **Create new** and enter "RG_Web_Server" as the name of the new resource group.
3. Name: Enter "Web_Server" as the name of the virtual network.
4. Region: Choose the region that is closest to you.



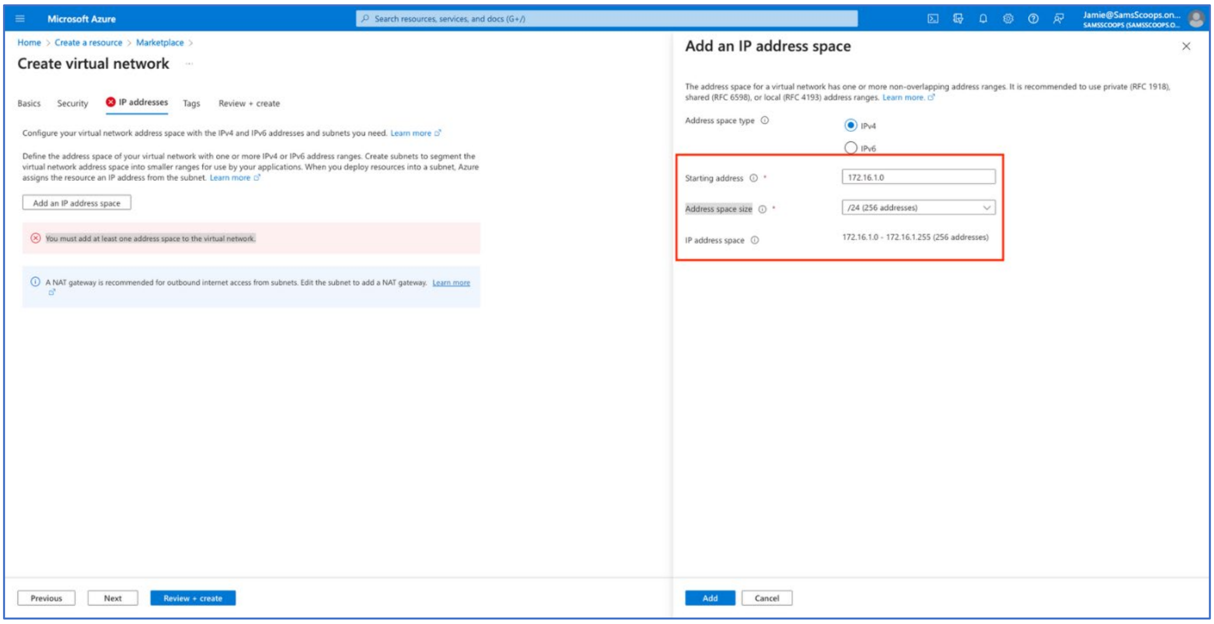


Step 5

1. Select **IP addresses**.



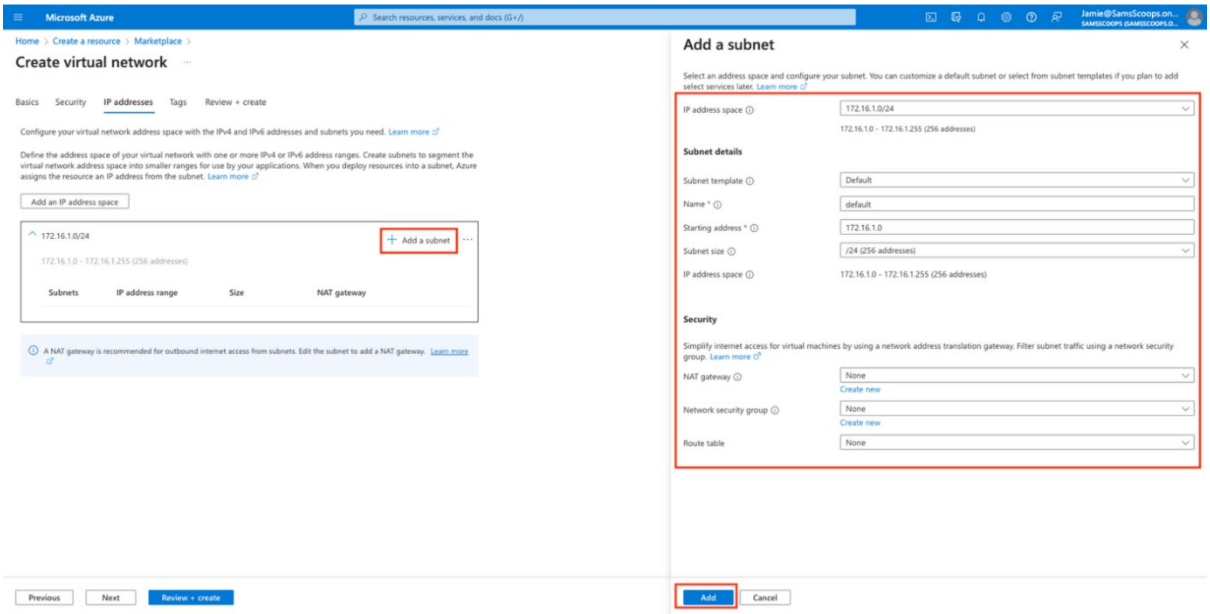
2. Delete the default address space by selecting the dots (**...**) next to **Add a Subnet** and select **Delete Address Space**.
3. There is a warning: **You must add at least one address space to the virtual network**.
4. Select **Add an Address Space**.
5. Fill in starting address: **172.16.1.0**.
6. Fill in address space size: **/24 (256 Addresses)**.
7. Select **Add**.



Step 6

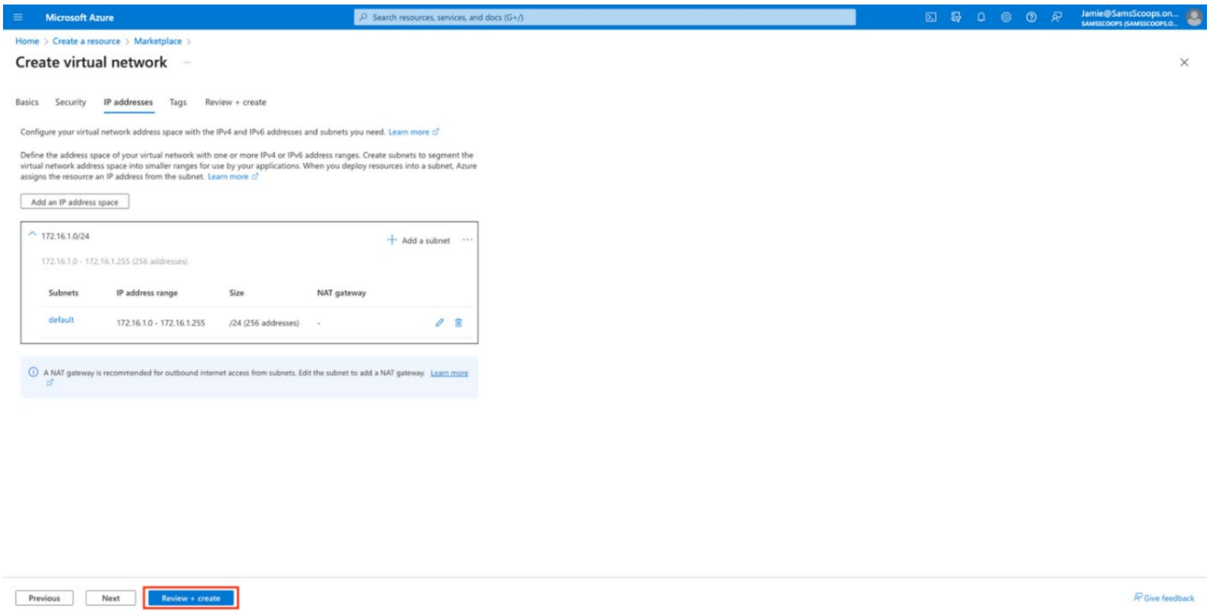
Now, a Subnet has to be added.

1. Select **+ Add a subnet**.
2. Leave the details as default and select **Add**.



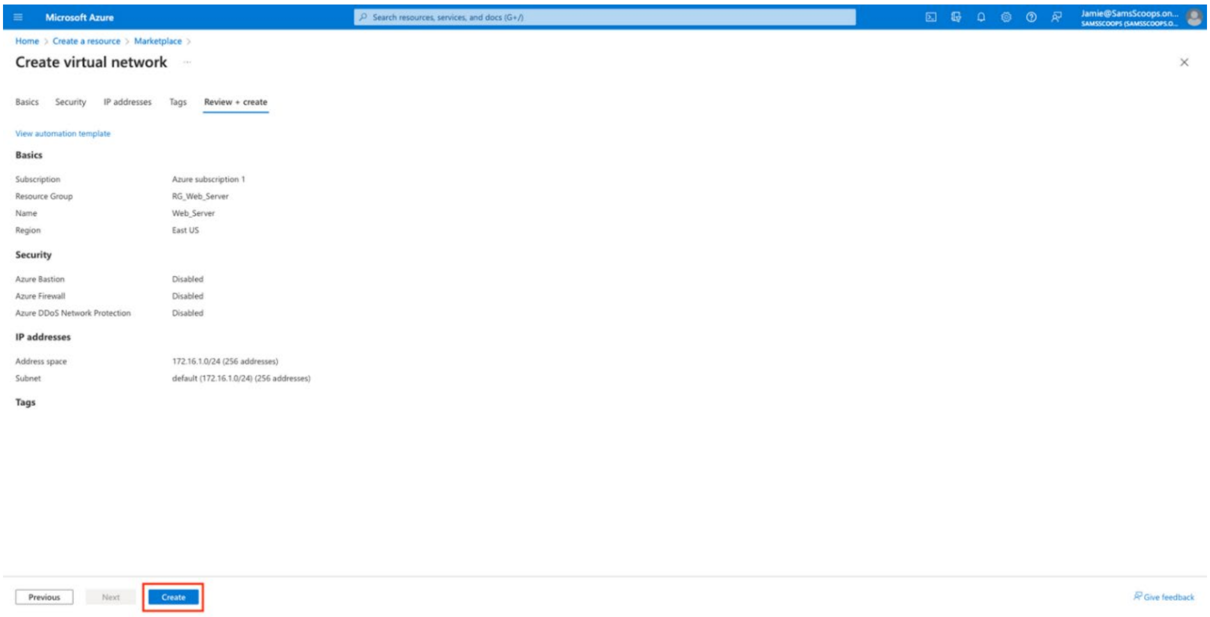
Step 7

Select the **Review+ create** button to review the settings.

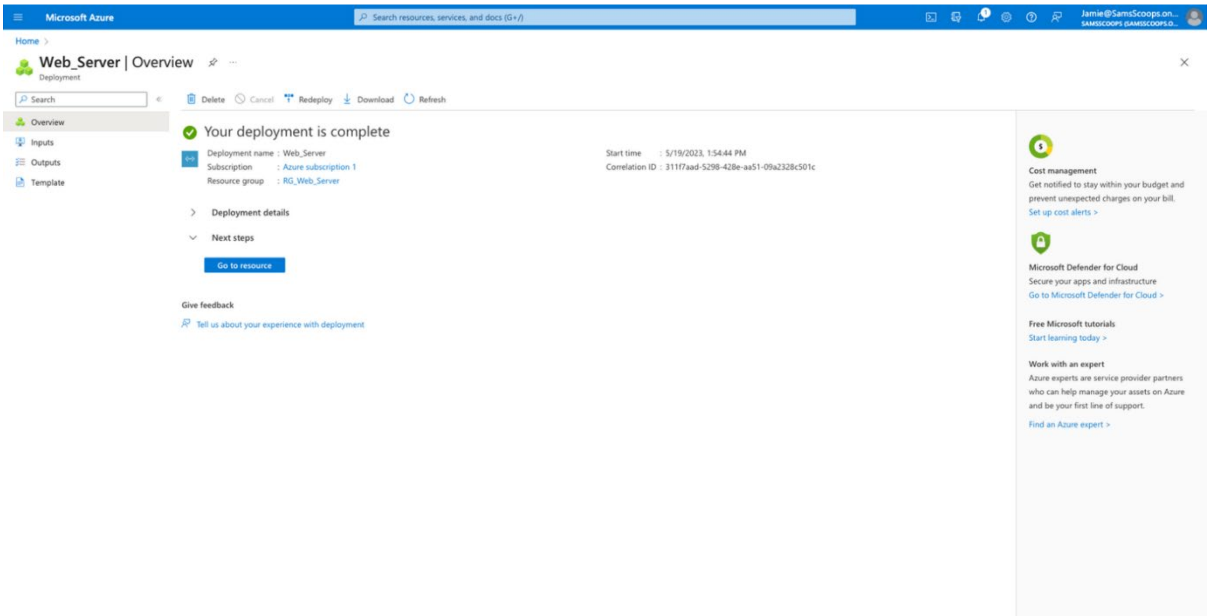


Step 8

Select the **Create** button to create the virtual network.



At this stage there should be a notification that says **Deployment in progress**. When the deployment is completed, it should say **Your deployment is complete**.



Once the virtual network is created, you can proceed with creating the virtual machine for the Sam's Scoops web server.

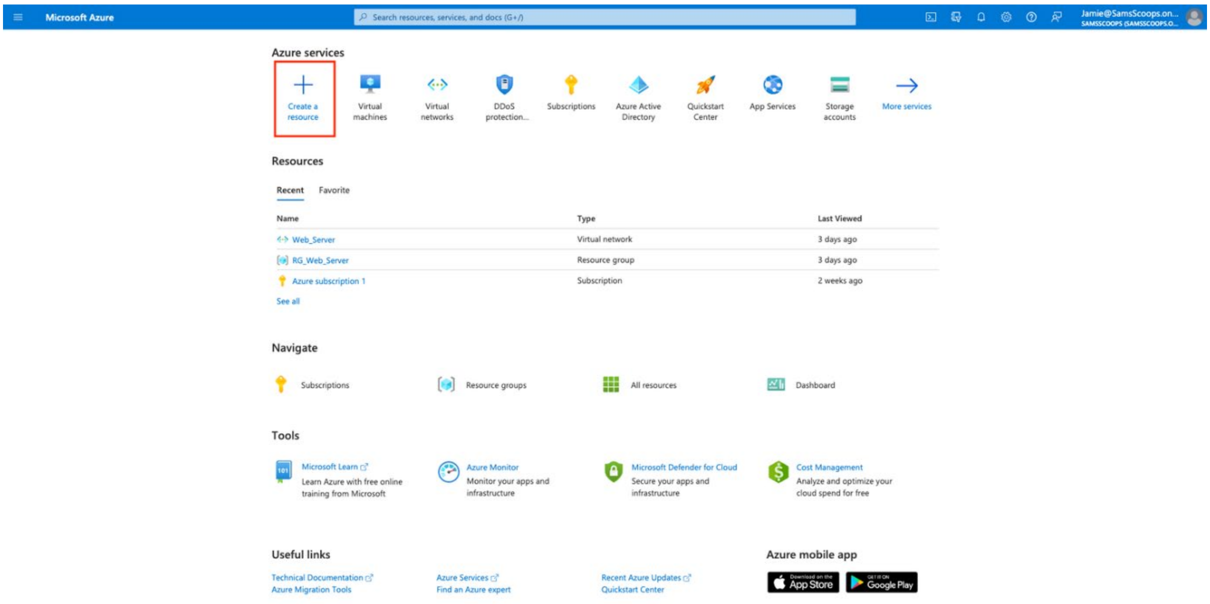
Create the virtual machine

Step 1

Sign in to the [Azure portal](https://portal.azure.com) with your credentials.

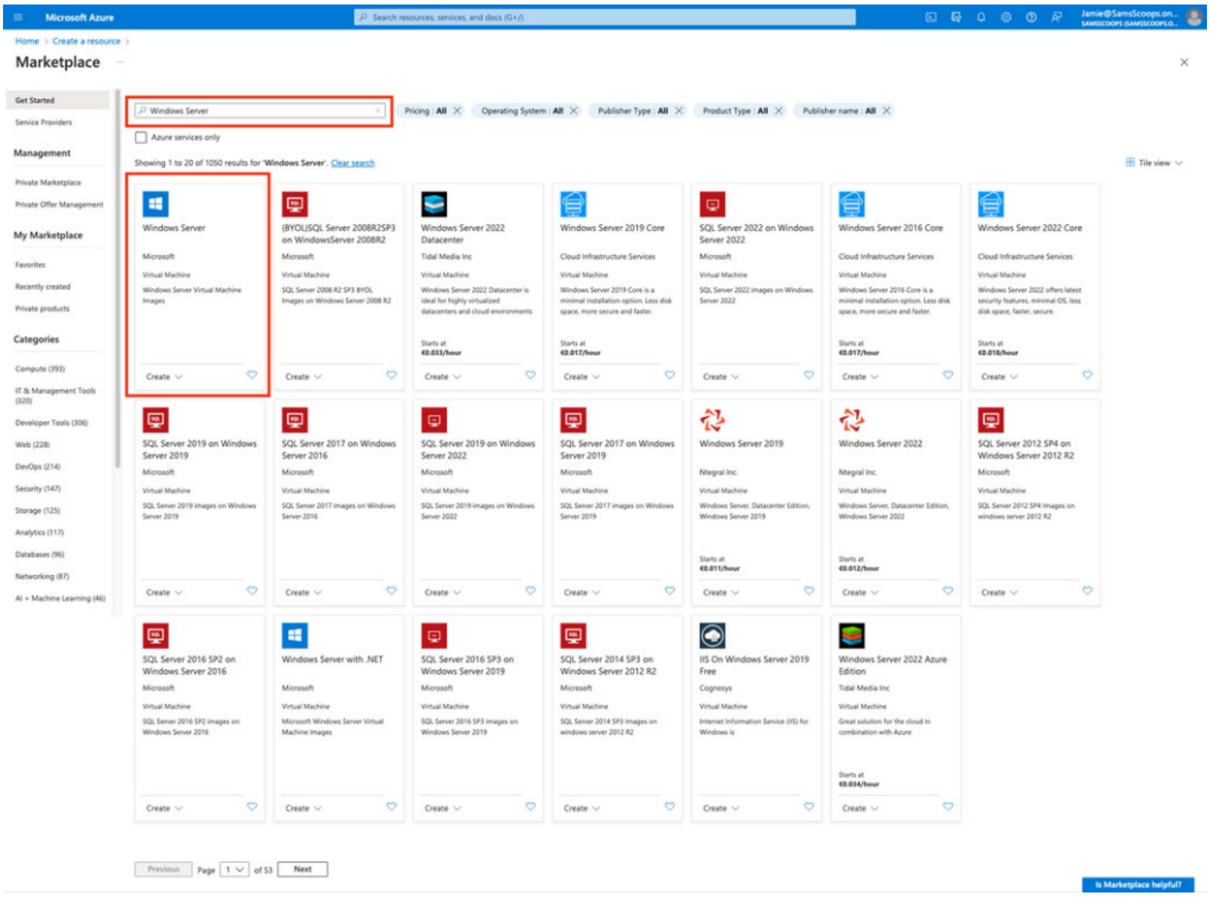
Step 2

In the Azure portal menu, select the **Create a resource** button located on the left-hand side of the screen.

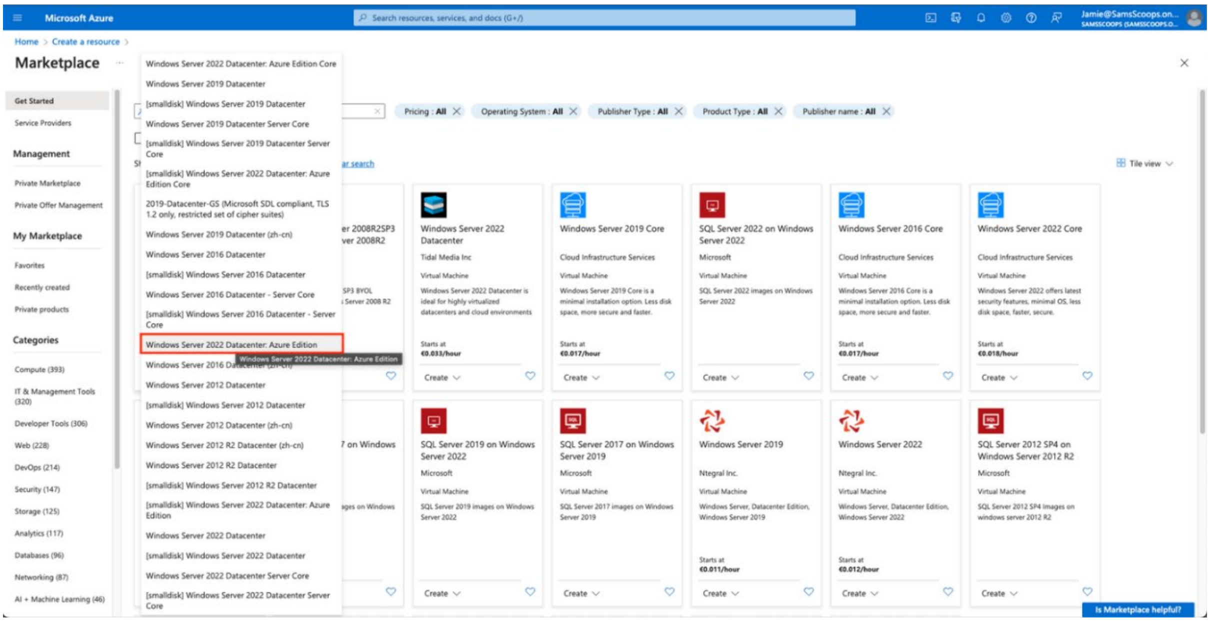


Step 3

Search for "Windows Server" in the search bar and select **Windows Server** from the results.



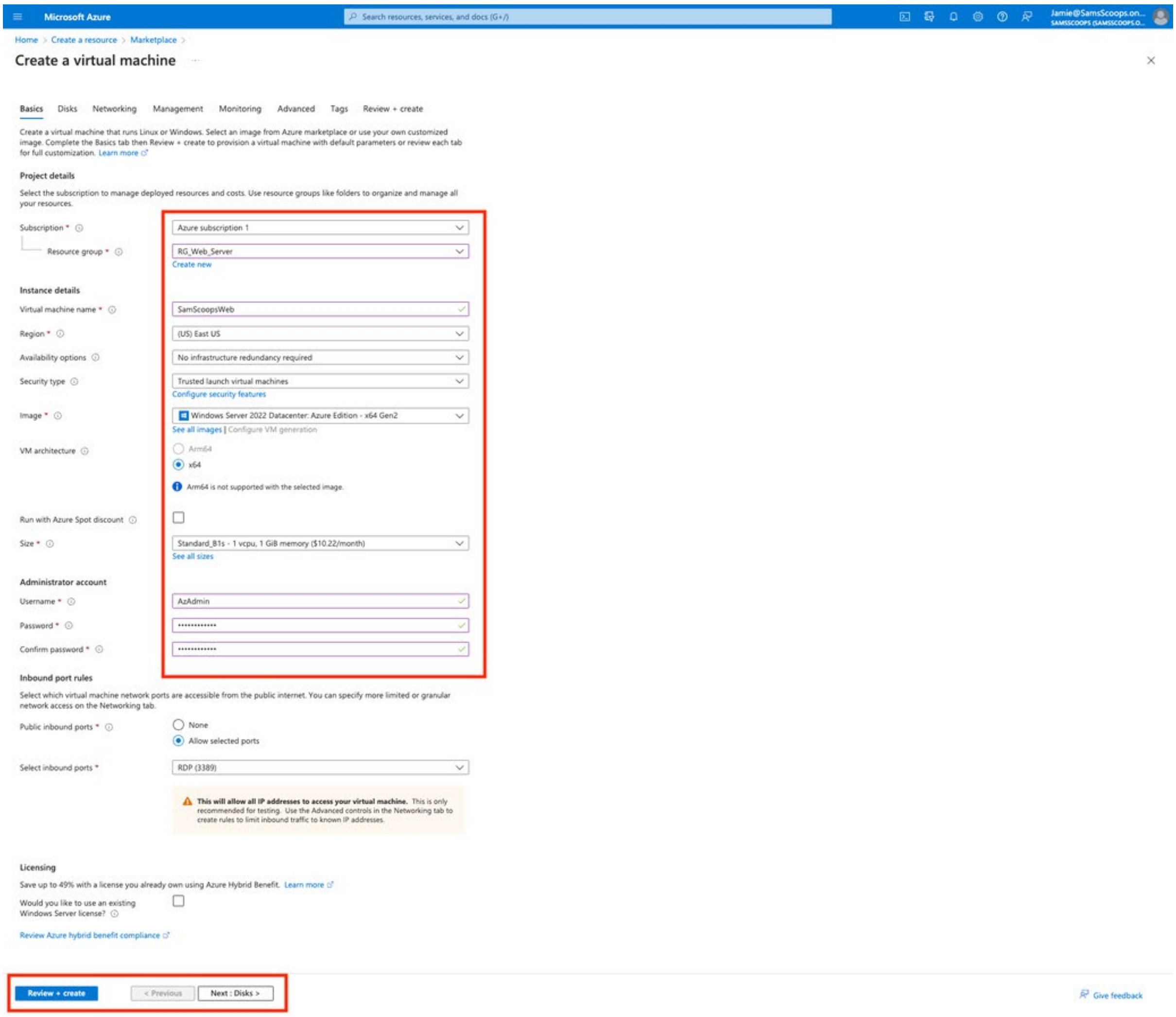
1. Select the **Create** and select **Windows Server 2022 Datacenter: Azure Edition**.



Step 4

In the **Basics** tab of the Create a virtual machine wizard, fill out the following information:

1. Subscription: Choose the subscription that you want to use.
2. Resource group: An existing resource group, **RG_Web_Server**.
3. Virtual machine name: Enter "SamScoopsWeb".
4. Region: Choose the region that is closest to you.
5. Image: **Windows Server 2022 Datacenter –x64 Gen2**.
6. Size: An appropriate size for your virtual machine. It will be good idea to use choose the cheapest option for this exercise.
7. Username: **AzAdmin**
8. Password: **P@$$@1234567**
9. Confirm Password: **P@$$@1234567**

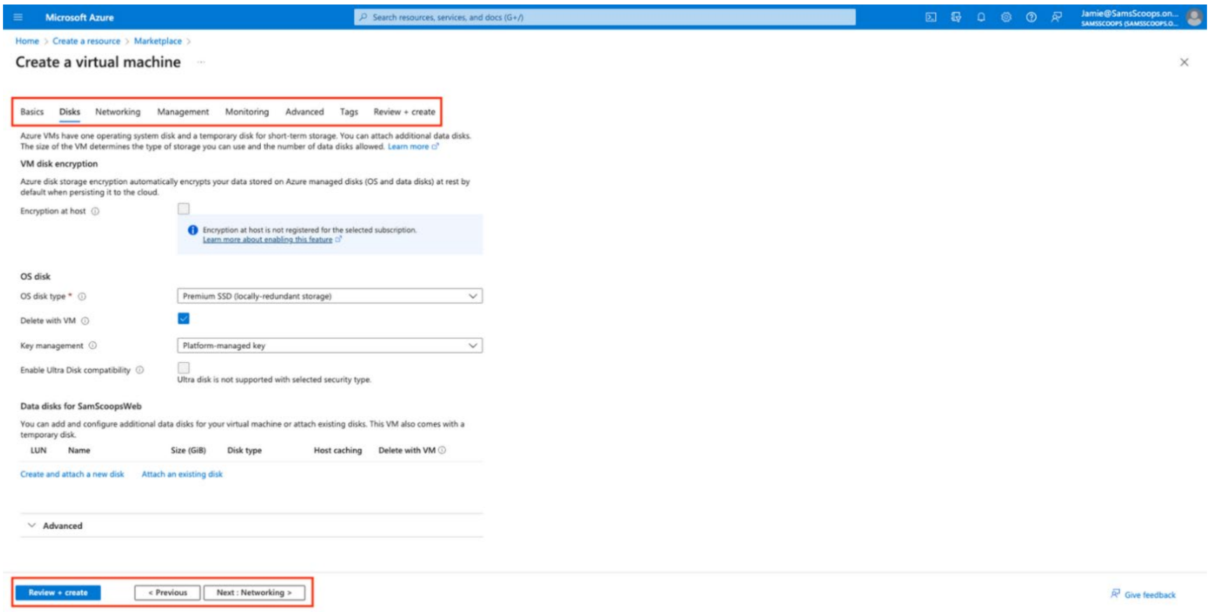


Step 5

Select the **Next: Disks** button to proceed to the Disks tab.

Step 6

Leave the default settings for OS disk and select the **Next: Networking** button to proceed to the **Networking** tab.



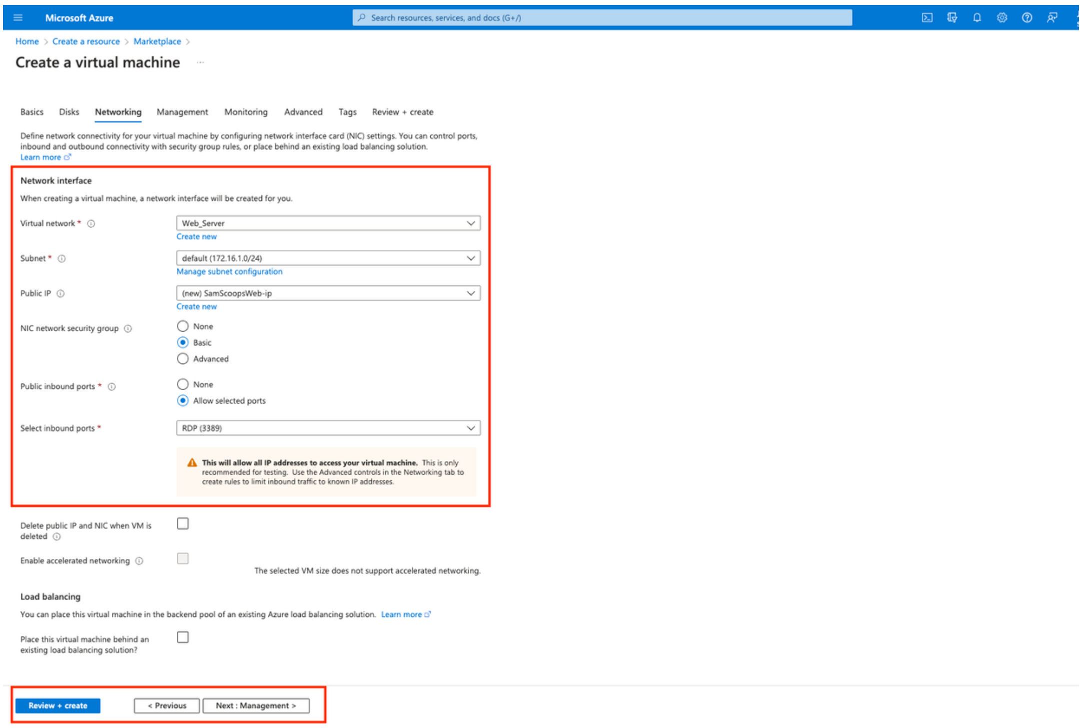
Step 7

In the **Networking** tab, select the following settings:

1. Virtual network: **Web_Server**.
2. Subnet: **default.**
3. Public IP: **Create new** and enter a name for the new public IP address.
4. NIC network security group: Leave the default setting.

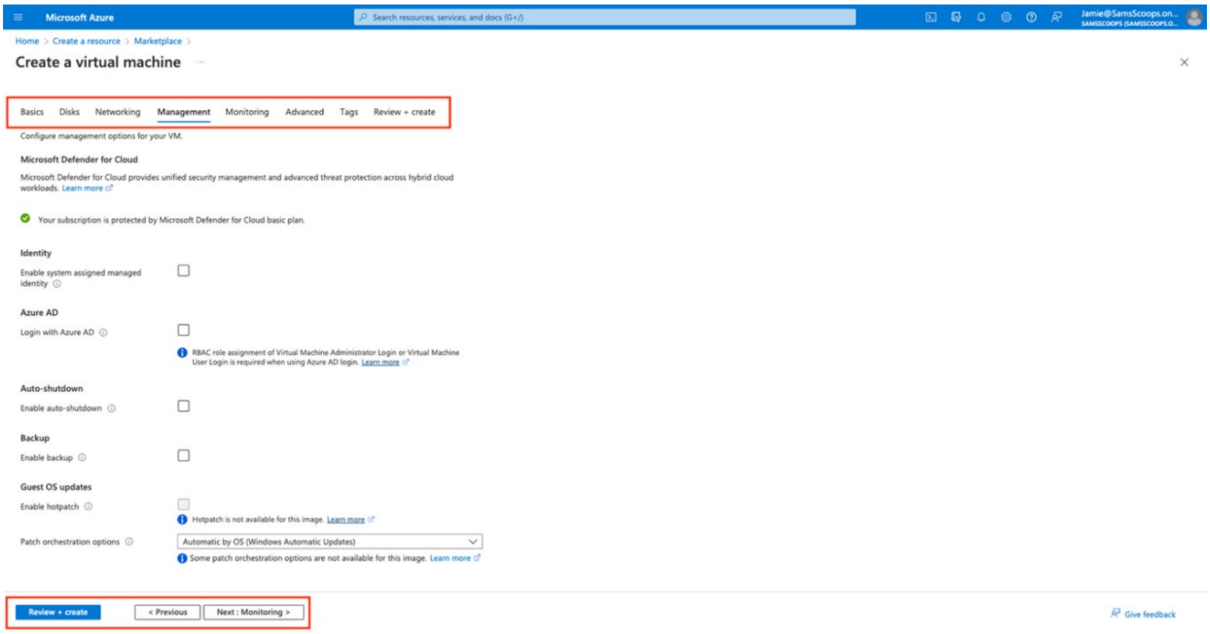
Step 8

Select the **Next: Management** button to proceed to the Management tab.



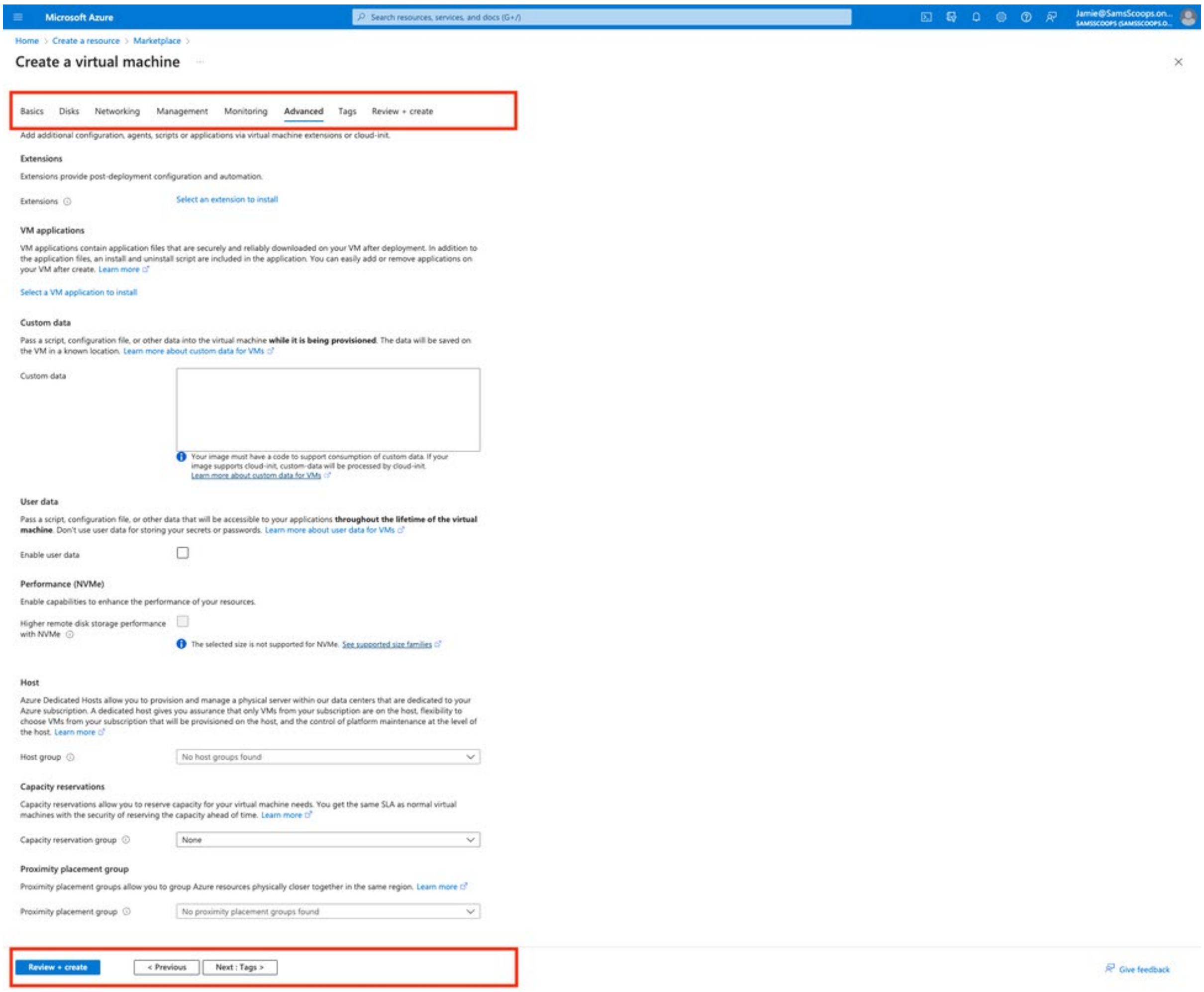
Step 9

In the **Management** tab, leave the default settings and select the **Next: Monitoring** button to proceed to the **Monitoring** tab. Leave everything as default. Then select the **Next: Advanced** button to proceed to the **Advanced** tab.



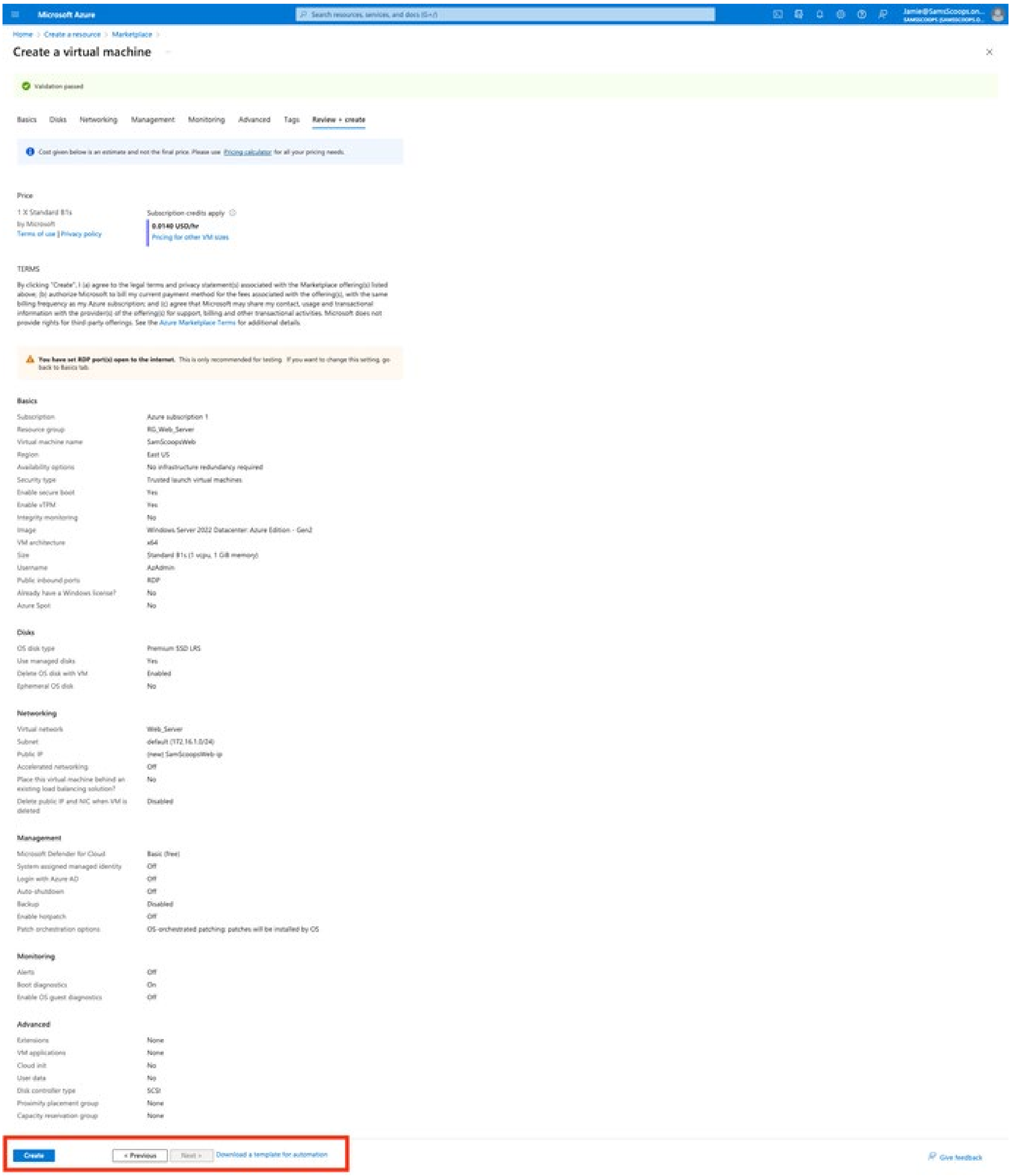
Step 10

In the **Advanced** tab, leave the default settings and select the **Review + create** button.



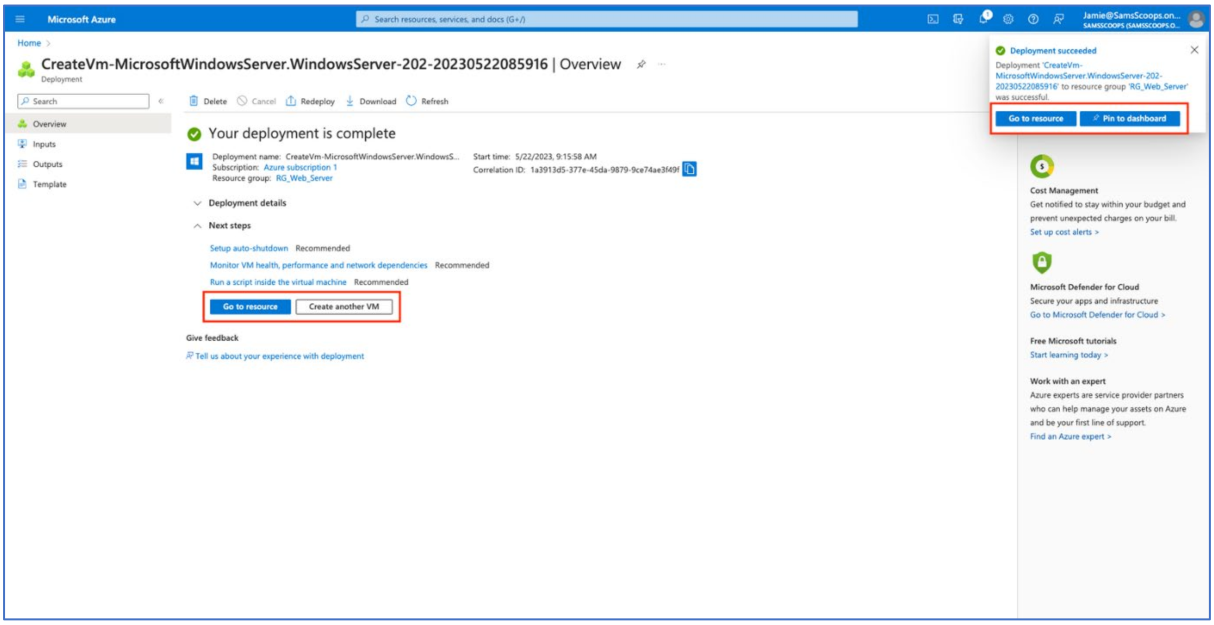
Step 11

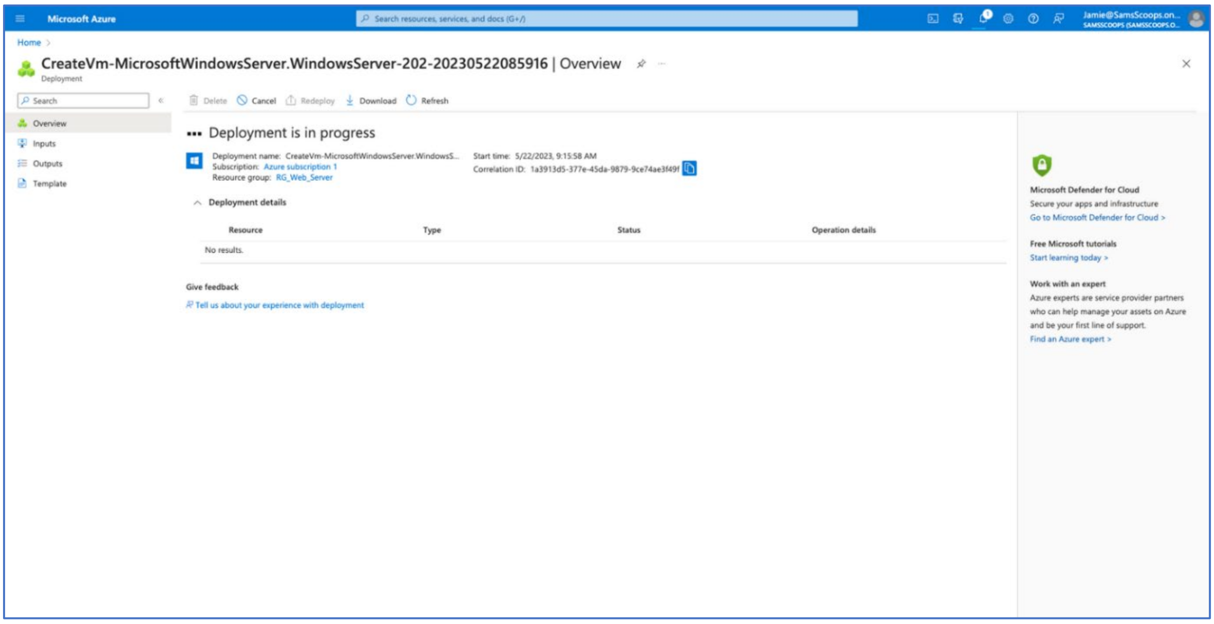
Review the settings for your virtual machine and select the **Create** button to start the deployment of your virtual machine and wait for deployment.



Step 12

Wait for the deployment to complete.



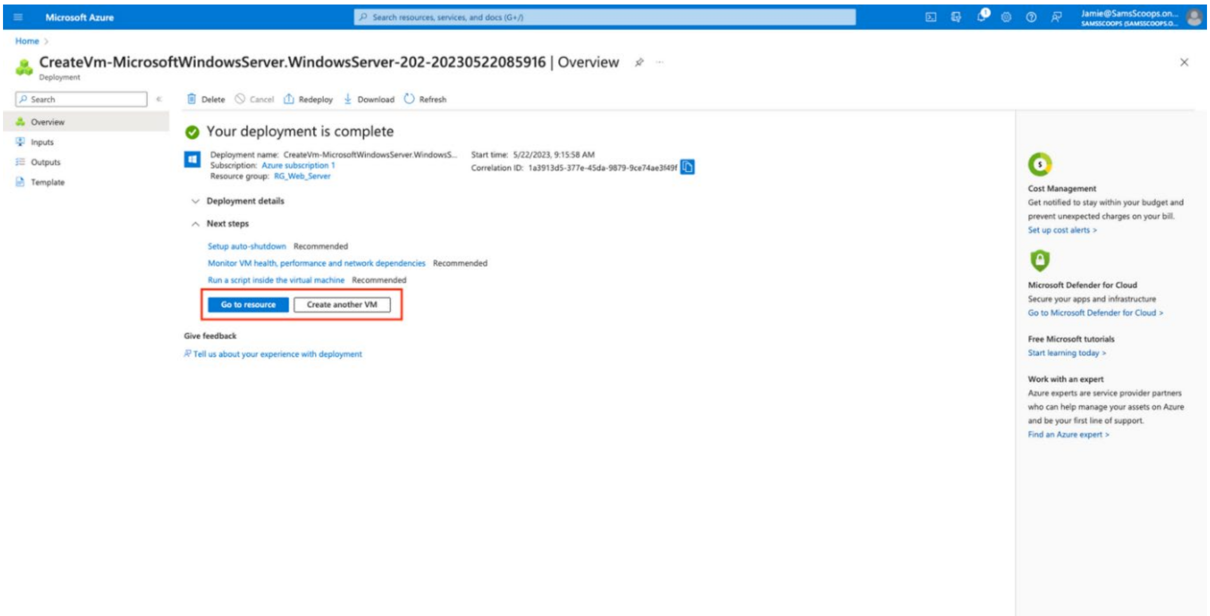


Well done. The virtual machine is now deployed. You are now ready for the next phase where you need to connect to the virtual machine.

Connect to virtual machine

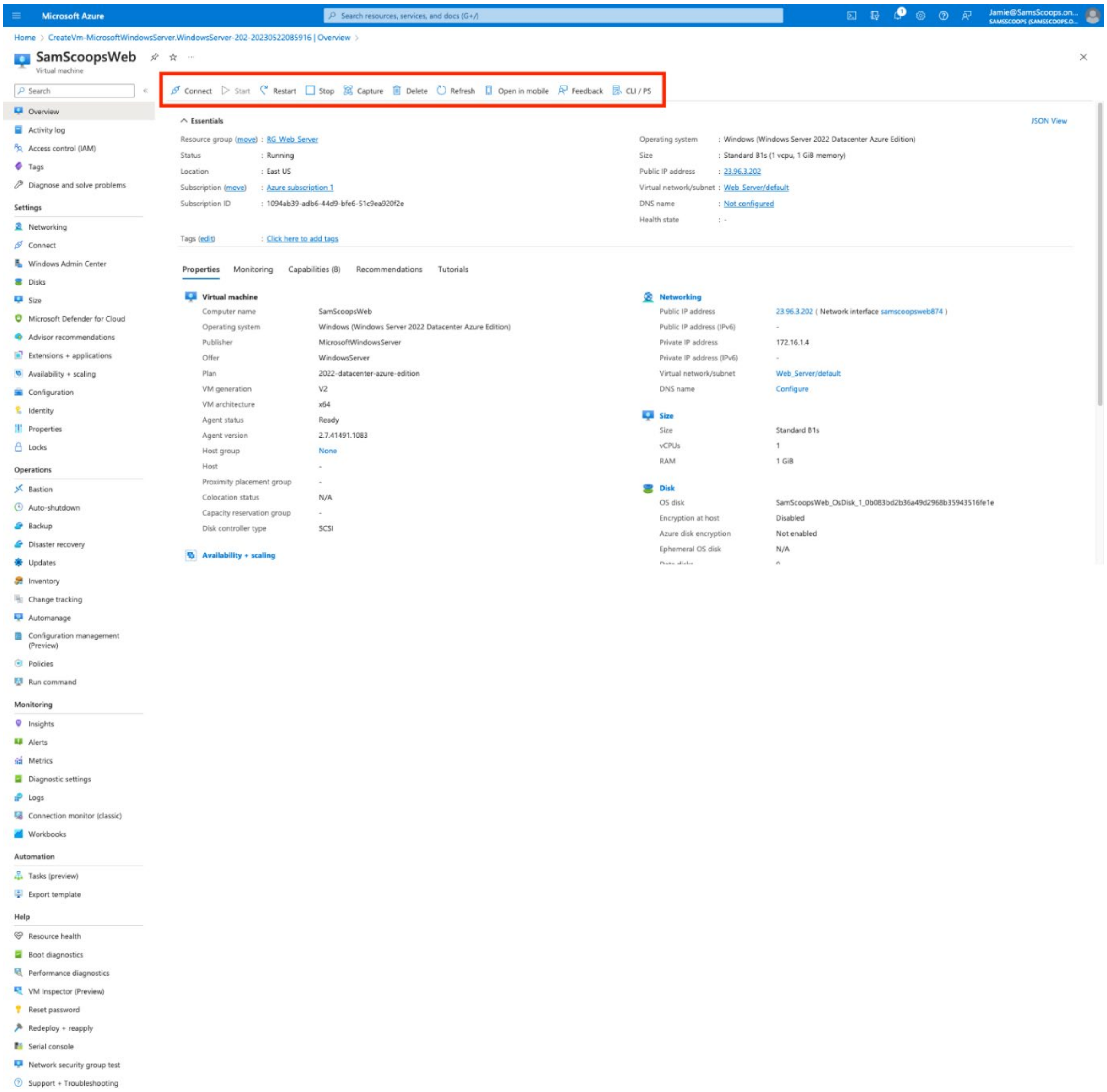
Step 1

Once the deployment is complete, select the **Go to resource** button to navigate to the virtual machine page.



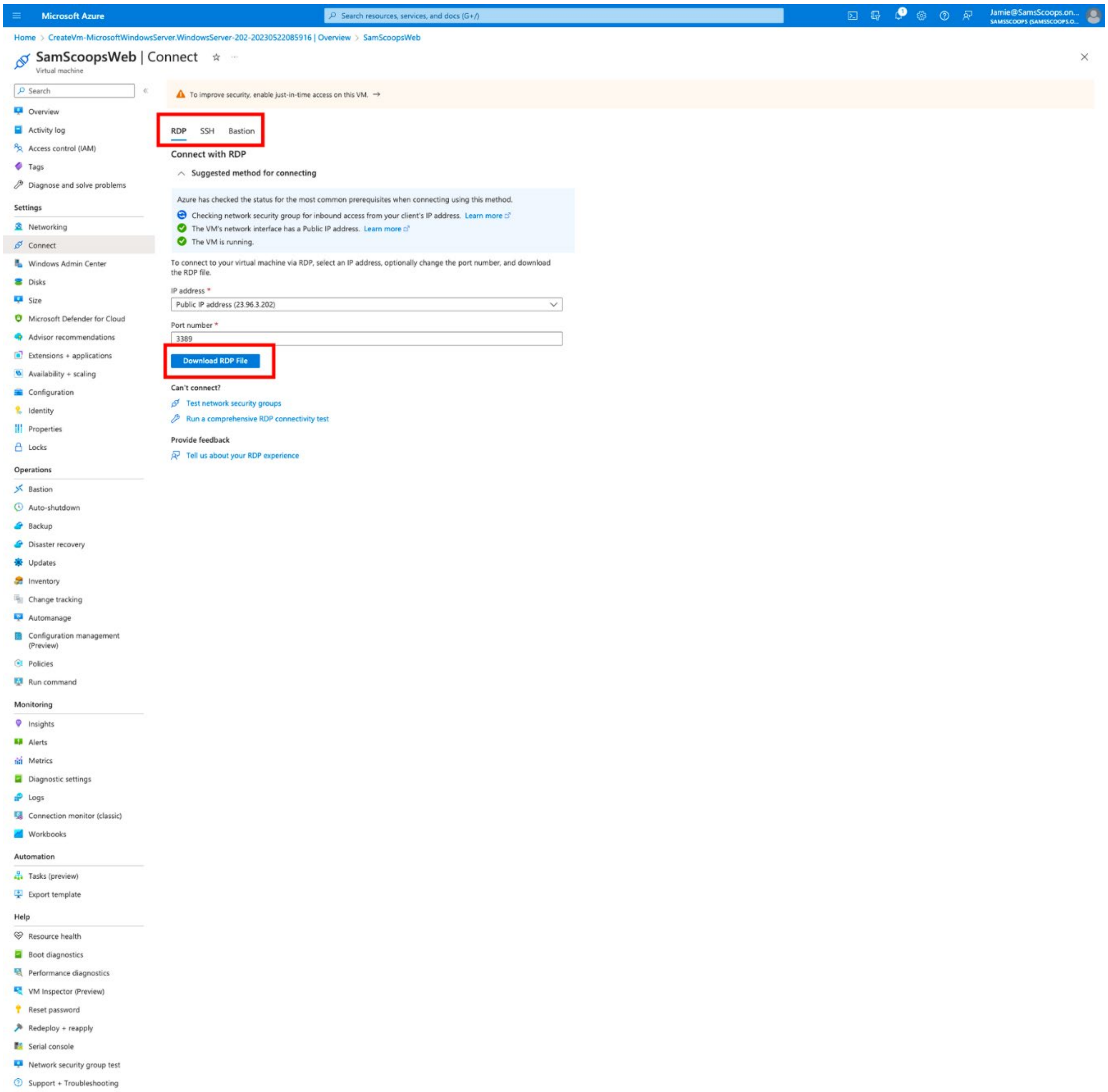
Step 2

On the virtual machine page, select the **Connect** button and **RDP** from the option menu to download the RDP file.



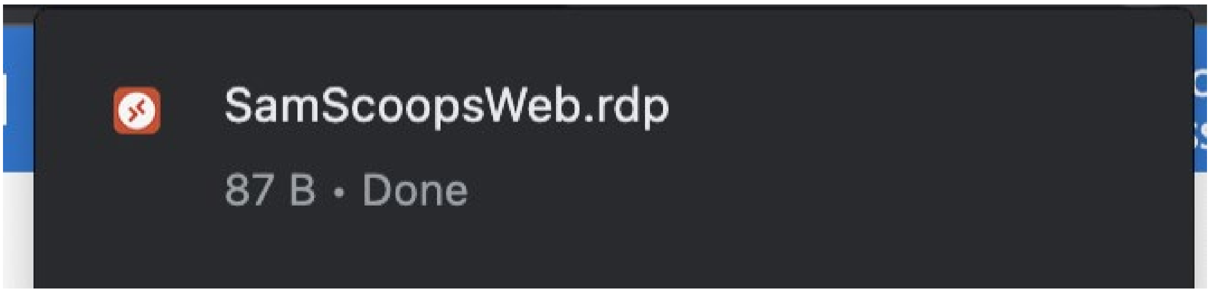
Step 3

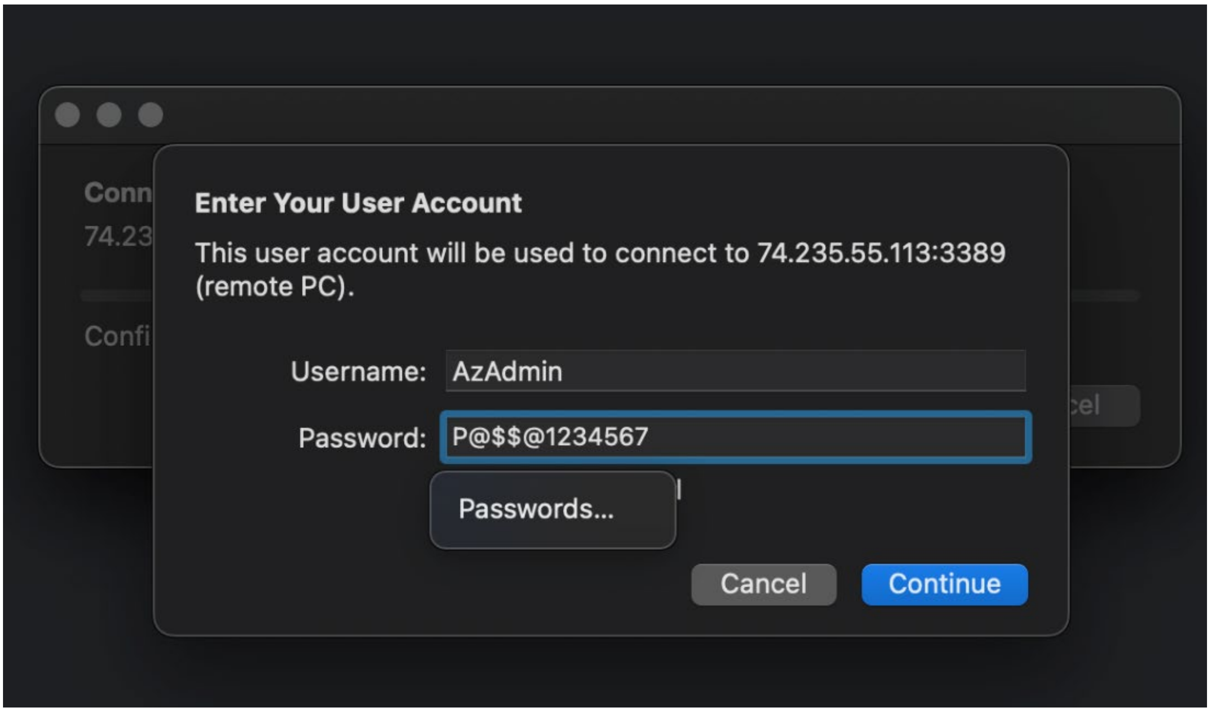
Select **RDP** from the options menu to download the RDP file. Select **Download RDP file**.



Step 4

Open the downloaded RDP file and connect by using the credentials you have set while creating the virtual machine to connect to the virtual machine.





You have now created a Windows virtual machine in the Azure portal for **SamScoopsWeb**.